

AYE
TECH HUB

• PROTASTRUCTURE • LESSON 02

Interface and Navigation

Master Every Zone, Tool and Shortcut in ProtaStructure

Lesson 02 of 30 | ProtaStructure Program

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- The ProtaStructure interface — 7 zones fully explained
- Main toolbar: every icon and what it does
- The Building Tree — navigating stories and structural elements
- 3D and 2D view navigation: orbit, pan, zoom and section views
- 20 essential keyboard shortcuts for professional workflow speed

The ProtaStructure Interface — 7 Zones

ProtaStructure has a clean, engineering-focused interface. Unlike general-purpose CAD software, every zone is purpose-built for structural design. Understanding what each zone does before you start modelling will save you hours of searching for tools.



Figure 1 — ProtaStructure Interface: 7 Key Zones

Zone	Location	Purpose	Most Used For
① Menu Bar	Very top	All commands organised into File, Edit, View, Define, Input, Analysis, Design, Reports	Accessing less-common commands; setting project codes and materials
② Main Toolbar	Below menu	One-click access to the most frequent actions as icon buttons	New/Open/Save, Undo/Redo, switching 2D/3D, launching analysis and design
③ Building Tree	Left panel	Hierarchical list of all stories (floors) and structural elements in the building	Navigating to a specific floor; selecting element types; managing loads
④ 3D/2D Viewport	Centre	Main modelling canvas — shows 2D plan or 3D model of the building	All structural element input, selection, editing and visual inspection
⑤ Properties / Results	Right panel	Shows parameters of the selected element; switches to design results after analysis	Editing element dimensions, material grades; reviewing design ratios
⑥ Story Selector	Top of viewport	Switch between floors; toggle 2D plan / 3D view; filter element visibility	Moving between Ground, 1st Floor, 2nd Floor, Roof while modelling
⑦ Output / Status Bar	Bottom	Displays analysis messages, warnings, design failures, and progress	Checking for warnings after analysis; confirming successful design completion

Always watch the Output Bar (Zone 7) after running analysis or design. If ProtaStructure shows red messages, elements have failed design checks. Yellow messages are warnings. Green messages confirm success. Never ignore the output bar — it is your quality control checkpoint.

The Building Tree — Your Project Navigator

The Building Tree is the most important navigation tool in ProtaStructure. It displays the complete hierarchical structure of your project — from the top-level Project down to individual structural elements on each floor. Everything you model is organised here automatically.

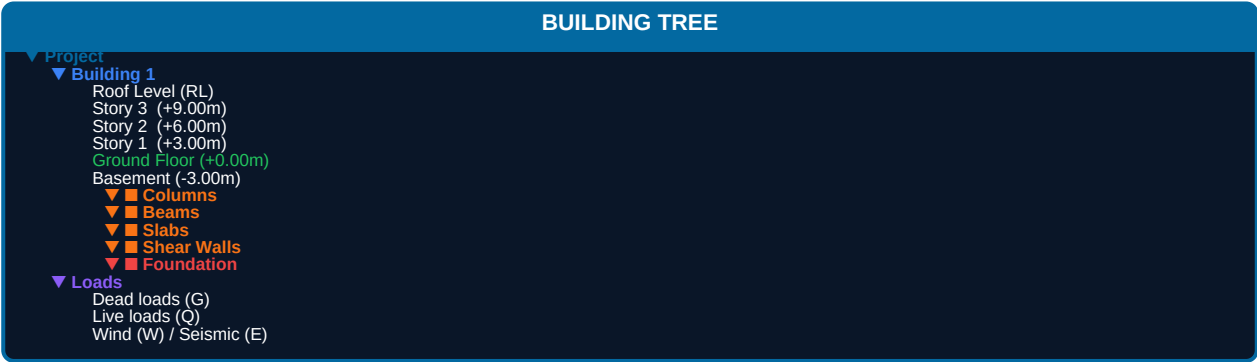


Figure 2 — ProtaStructure Building Tree: Project → Stories → Elements

Tree Level	What It Contains	Key Actions
Project	Top level — name, code, material standards	Right-click → Project Properties to set design codes, seismic data, soil parameters
Building	Contains all floors and the global structural system	Right-click → Add Story to create new floors; set building dimensions
Story (Floor level)	Each individual floor with its elevation (e.g. Ground Floor +0.00m)	Double-click to activate that floor for editing; right-click to add/delete/rename
Element type folders	Columns, Beams, Slabs, Shear Walls, Foundations grouped per floor	Click folder to show/hide all elements of that type; right-click to select all
Individual elements	Each specific beam, column, slab with its ID number and cross-section	Click to select; double-click to open properties dialog for editing
Loads	Dead loads, live loads, wind, seismic — applied to whole building or per floor	Expand to review all applied loads; right-click to modify load values

Main Toolbar — Icon Guide

Icon Group	Icons	Function
File Operations	New · Open · Save · Save As · Print	Basic file management — Ctrl+S to save frequently
Edit Operations	Undo · Redo · Cut · Copy · Paste · Delete	Undo (Ctrl+Z) is unlimited in ProtaStructure — use it freely
View Controls	2D Plan · 3D View · Front · Side · Top · Isometric	Rapidly switch between views without going to the menu
Zoom Controls	Zoom In · Zoom Out · Zoom Fit · Zoom Window	F = Fit all; scroll wheel = zoom in/out at cursor
Analysis & Design	Run Analysis · Run Design · Check · Optimise · Stop	The most-used buttons — run analysis first, then design
Display Options	Show axes · Show loads · Show results · Mesh · Deformed shape	Toggle overlays on and off to keep the view readable

3D and 2D Navigation

ProtaStructure uses a standard 3D engineering viewport. The 2D plan view is used for element input and editing. The 3D view is used for visual inspection, load review, and checking results. You switch between them using the Story Selector toolbar or keyboard shortcuts.

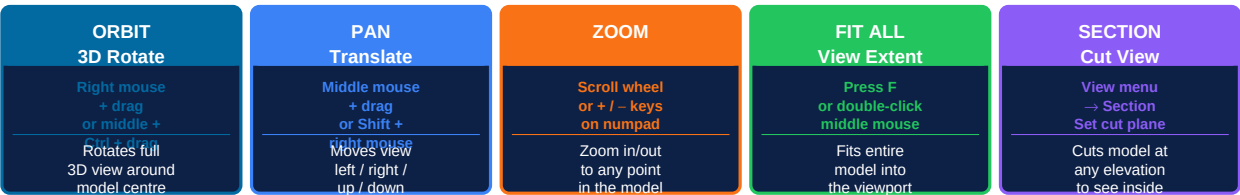


Figure 3 — ProtaStructure 3D and 2D View Navigation Controls

View Mode	Activated By	Best Used For
2D Plan View (per story)	Story Selector → select floor → "2D" button	Drawing walls, beams, columns, slabs; placing loads; reviewing plan geometry
3D Perspective View	Toolbar "3D" button or press 3	Checking model completeness; reviewing frame geometry; presenting to clients
3D Wire Frame	Display Options → Wire Frame	Faster rendering for large models; seeing through slabs to check structure below
Elevation/Section View	View menu → Section → Define cut plane	Checking column heights; reviewing beam-column connections; verifying levels
Deformed Shape View	After analysis → Results toolbar → Deformed	Verifying structural behaviour; checking maximum deflection location
Force Diagram View	After analysis → Results → Bending Moment / Shear Force	Reviewing analysis results; identifying critical sections for design

Essential ProtaStructure Keyboard Shortcuts

Shortcut	Action	Shortcut	Action
Ctrl+N	New project	Ctrl+O	Open project
Ctrl+S	Save project	Ctrl+Z	Undo
Ctrl+Y	Redo	Delete	Delete selected element
F	Zoom to fit all	Esc	Deselect / cancel
1	Switch to 2D plan view	3	Switch to 3D view
Ctrl+A	Select all visible	Ctrl+G	Group elements
F5	Run analysis	F6	Run design
F7	Design check report	F8	Quantity report
Ctrl+R	Rotate selected element	Ctrl+M	Mirror element
Ctrl+C/V	Copy / Paste element	Ctrl+D	Duplicate to another story

Common Mistake	Consequence	Prevention
Modelling on the wrong story	Elements placed on wrong floor level — wrong heights in analysis	Always check the Story Selector before inputting any element

Not setting design codes before modelling	Analysis uses default or wrong codes; results are non-compliant	Project Properties → Design Codes → set EC2/ACI/BS before first element
Ignoring analysis warnings	Design results are unreliable; may contain failed elements not noticed	Always read the Output Bar after every analysis and design run
Not saving after analysis	Analysis results lost if program crashes or is closed	Ctrl+S immediately after successful analysis and design completion
Working only in 3D — not checking 2D plans	Element positions may appear correct in 3D but be offset from grid in plan	Always verify element positions in 2D plan view after 3D input

What comes next — PROTA-003: Structural Design Principles — the engineering fundamentals every ProtaStructure user must know: load types, load combinations (EC2/ACI), tributary areas, frame behaviour, and how ProtaStructure translates these into the analysis model.